(g) $n(T) = 0$

(h) T^{-1} exists.

Proof:- Given that $T : U \longrightarrow V$ is a linear map.

$\dim(U) = \dim(V) = p$.

(a) $\Rightarrow$ (b)

Let T is non-singular (an isomorphism)

$\Rightarrow$ T is one-one and onto [By def^n of non-singular]

$\Rightarrow$ T is one-one.

(b) $\Rightarrow$ (c)

Let T is one-one.

Let $u_1, u_2, \ldots, u_p$ are LI vectors of U.

$\Rightarrow T(u_1), T(u_2), \ldots, T(u_p)$ are LI vectors of V [By prev. theorem]

$\therefore$ T transforms LI subsets of U into LI subsets of V.

(c) $\Rightarrow$ (d)

Let T transforms LI subsets of U into LI subsets of V.

Now, let $\{u_1, u_2, \ldots, u_p\}$ is a basis for U.

Then $\{T(u_1), T(u_2), \ldots, T(u_p)\}$ is LI.

But $\dim(V) = \dim(U) = p$

So $\{T(u_1), T(u_2), \ldots, T(u_p)\}$ is a basis for V.

(d) $\Rightarrow$ (e)

Let T transforms every basis for U into a basis for V

Let $\{u_1, u_2, \ldots, u_p\}$ is a basis for U. Then

$\{T(u_1), T(u_2), \ldots, T(u_p)\}$ is a basis for V.

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